

The Math Behind Western Standoffs: Bayesian Games

Shaon Anwar

1 Introduction

In the late 1960s, following the explosive development of game theory, Hungarian-American economist John Harsanyi popularized the analysis of situations involving “Bayesian players” distinguished by their lack of complete knowledge of the game. These scenarios, coined “Bayesian games” became incredibly relevant to economic sciences because of their applicability to real-life situations; a model where each player has their own limited set of information avoids many of the unrealistic assumptions taken during a complete-information game.

It is worth noting that in a Bayesian scenario, as with a complete-information game, there exists an optimal solution concept for each player. A common non-Bayesian game solution concept is the Nash equilibrium, where each player cannot benefit from altering their strategy. Likewise, a Bayesian Nash equilibrium solution concept exists for Bayesian games. Since Bayesian players are forced to make assumptions regarding private information, an element of probability is introduced: each player’s strategy is determined by the probabilities of the other player’s potential roles.

2 The Bayesian Game

In game theory, a Bayesian game is a situation defined by the fact that its players hold incomplete information. In other words, players may have access to private information regarding the game, but lack the common knowledge of a complete-information game. To best illustrate this concept of incomplete information, let’s turn to a common example of a Bayesian game: “sheriff’s dilemma.”

In “sheriff’s dilemma,” let there be two players, A and B . Player B assumes the role of the sheriff, while player A assumes the role of either a criminal or an innocent civilian. Importantly, A is aware of their own role as either a criminal or a civilian and the other player’s status as the sheriff. B , on the other hand, is unsure whether or not A is a criminal or a civilian. Thus, B ’s knowledge of their own role is private information, a defining characteristic of Bayesian games.

Assume B decides to confront A , and each player must decide between shooting and not shooting the other. To further establish the players’ intentions, assume that if player A shoots, then player B , the sheriff, prefers to shoot back. Likewise, if A does not shoot, B prefers to not shoot either. On the other hand, if A is a criminal, they prefer to shoot in any case, and if A is a civilian, they prefer to not shoot in any case. Given that p is the probability of player A being a criminal and $1 - p$ is the probability of A being a civilian, we can deduce the scenario that represents Bayesian equilibrium by using expected payoffs and probability.

Each player's expected payoff in each scenario can be modeled by matrices; the upper matrix, C , assumes player A is a civilian, while the lower matrix, D assumes A is a criminal. In each cell, the first number represents player A 's expected payoff, while the second represents that of player B . The headings in the first column correspond to A 's actions, while the headings in the first row correspond to B 's actions. The expected payoffs in each cell are determined by the players' preferences in each scenario.

C	Shoot	Not Shoot
Shoot	-3, -1	-1, -2
Not Shoot	-2, -1	0,0

D	Shoot	Not Shoot
Shoot	0, 0	2, -2
Not Shoot	-2, -1	-1,1

Figure 1: Source: Joonas Vatto, Harvard University

2.1 Matrix C : A is a Civilian

In matrix C , where player A is a civilian, A most prefers to not shoot and not be shot, which would yield a designated expected payoff of 0. A 's next most optimal scenario is to shoot and to not be shot, to which we will assign an expected payoff of -1 . A then prefers to not shoot and be shot, to which we will assign -2 . Finally, A 's worst case scenario is shooting and being shot, yielding an expected payoff of -3 .

As for player B , the optimal scenario is to not shoot in response to not being shot, yielding an expected payoff of 0. B then equally prefers shooting in response to being shot, and shooting in response to not being shot (recall that B does not know if A is a criminal or civilian), with a payoff of -1 . B 's worst-case scenario is being shot without shooting back, with a payoff of -2 .

2.2 Matrix D : A is a Criminal

In matrix D , player A is a criminal, and thus prefers to shoot and not be shot with a payoff of 2. If both parties shoot, B has an expected payoff of 0, which is preferred to neither party shooting (payoff of -1) and being shot at without shooting (payoff of -2).

As the sheriff, B 's preferences generally stay the same, as they do not know if A is a criminal or a civilian. However, if neither party shoots, B receives a payoff of 1, since an optimal scenario results in an arrest after the confrontation.

3 The Optimal Strategies

As demonstrated by the matrices' expected payoffs, A 's strategy simply hinges on their role. If A is a criminal, they should shoot regardless, and if A is a civilian, they should not shoot.

B 's path is naturally reliant on the chances of A being a criminal. Recall that A 's probability of being a criminal is p . We can gauge the expected value by finding the sum of the products of the probability of each case and B 's expected payoffs:

- If the sheriff shoots: $((1 - p) \cdot -1) + (p \cdot 0) = p - 1$
- If the sheriff does not shoot: $((1 - p) \cdot 0) + (p \cdot -2) = -2p$.

Given that the sheriff will shoot if the expected value of shooting outweighs the expected value of not shooting, we can deduce that the sheriff should shoot if $p - 1 > -2p$. After isolating p , we can finally conclude the sheriff shoots when $p > \frac{1}{3}$.

References

- Acemoglu, Daron, and Asu Ozdaglar. "Incomplete Information: Bayesian Nash Equilibria, Auctions ...". MIT Economics, November 23, 2009.
[https://economics.mit.edu/sites/default/files/inline-files/Lectures 19-21 - Bayesian Nash Equilibria, Auctions and Introduction to Social Learning.pdf](https://economics.mit.edu/sites/default/files/inline-files/Lectures%2019-21%20-%20Bayesian%20Nash%20Equilibria,%20Auctions%20and%20Introduction%20to%20Social%20Learning.pdf).
- Hayes, Adam. "Bayes' Theorem: What It Is, the Formula, and Examples." Investopedia, March 30, 2024.
<https://www.investopedia.com/terms/b/bayes-theorem.asp>.
- Obara, Ichiro. "Bayesian Nash Equilibrium." UCLA Economics, February 1, 2012.
<http://www.econ.ucla.edu/iobara/BNE201B.pdf>.
- Vättö, Joonas. Finding Bayes-Nash Equilibria, October 2, 2019.
<https://canvas.harvard.edu/courses/61567/files/8597692/download?verifier=drBHWx30nYWvYVXXzkcahrmfaojZhBiKPQ5y9KTz>.